🔑 **Answer Key — Teacher Copy · Lesson 5-3**

MSTAR PRACTICE WORKSHEET

Lesson 5-3: Determine the Area of Trapezoids

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

A. 24 sq cm

B. 32 sq cm

C. 40 sq cm

D. 64 sq cm

$A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32$ sq cm.

Question 2 SELECTED RESPONSE — CORRECT: B

A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

A. 3 ft

B. 5 ft

C. 9 ft

D. 10 ft

$A = \frac{1}{2} \times (b_1 + b_2) \times h \rightarrow 45 = \frac{1}{2} \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5$ ft.

Question 3 SELECTED RESPONSE — CORRECT: C

What is the first step when finding the area of a trapezoid?

A. Multiply the two bases

B. Subtract the shorter base from the longer base

C. Add the two bases together

D. Divide the height by 2

The formula is $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The first step is to add the two parallel bases.

Question 4 SELECTED RESPONSE — CORRECT: C

A trapezoid has bases of 5 in and 9 in, and a height of 6 in. What is its area?

A. 27 sq in

B. 30 sq in

C. 42 sq in

D. 84 sq in

$A = \frac{1}{2} \times (5 + 9) \times 6 = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42$ sq in.

Question 5 SELECTED RESPONSE — CORRECT: B

Blueprint 1: A trapezoidal lobby window has bases of 6 ft and 10 ft and a height of 4 ft. What is its area for the glass order?

A. 16 sq ft

B. 32 sq ft

C. 40 sq ft

D. 60 sq ft

Use $A = \frac{1}{2} \times (b1 + b2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32$ sq ft.

Question 6 SELECTED RESPONSE — CORRECT: B

Blueprint 2: A trapezoidal skylight has bases of 9 ft and 7 ft and a height of 6 ft. What is its area?

A. 44 sq ft

B. 48 sq ft

C. 63 sq ft

D. 96 sq ft

A = ½ × (9 + 7) × 6 = ½ × 16 × 6 = ½ × 96 = 48 sq ft.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A trapezoidal window has parallel edges of 4 feet and 8 feet and a height of 5 feet. What is its area?

A. 17 square feet

B. 20 square feet

C. 30 square feet

D. 60 square feet

Area = ½ × (b_1 + b_2) × h = ½ × (4 + 8) × 5 = ½ × 12 × 5 = 30 square feet.

- **A:** That adds all three measurements. The rule multiplies the average of the parallel edges by the height.
- **B:** That is 4 times 5, using only one of the two parallel edges. Average them first.
- **D:** The halving step was skipped. 4 plus 8, times 5, is twice the area.

Part B — correct answer: A

What does the quantity (b_1 + b_2) ÷ 2 represent in the trapezoid area formula?

A. The average length of the two parallel bases.

B. The perimeter of the trapezoid.

C. The length of the slanted side.

D. The height of the trapezoid.

Averaging the two bases turns the trapezoid into a rectangle of the same area with that average width, which is then multiplied by the height.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about trapezoids and their area:

- **A** A trapezoid has exactly one pair of parallel sides called bases.
- **B** The height is measured perpendicular to both parallel bases.
- **C** The two bases must be equal in length.
- **D** A trapezoid can be split into two triangles to find its area.
- **E** Area = ½ × (b_1 + b_2) × h.

A, B, D, and E are true. C is false — if both bases were equal the figure would be a parallelogram.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 7 ft. Which shape has the greater area? Explain your reasoning using the formulas.

Model answer: The trapezoid's area is ½ × (5 + 11) × 8 = ½ × 16 × 8 = 64 sq ft and the parallelogram's is 7 × 8 = 56 sq ft, so the trapezoid is greater by 8 sq ft. Since both shapes share the height of 8 ft, the comparison really comes down to the trapezoid's average base, ½ × (5 + 11) = 8 ft, against the parallelogram's base of 7 ft.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.